

Praful J. Patil

📍 Pune ✉️ prafulp1008@gmail.com ☎️ +91 7620091008
in Praful patil

Profile Summary :

- I am an Embedded System Engineer Specialized in Embedded System Hardware & Software. I wish to build a successful career by utilizing and improving my skills and contribute in the success of my organization. I am experienced in following
- Embedded architecture design, Embedded C Device Drivers & Debugging..
- Circuit testing & troubleshooting.

Technical Qualifications:

Technology (S/W): Embedded C Development, IOT-based development.

Micro-Processors/Controllers (Hands-on): ARM Cortex M0 ,M4,M4F, series,8051 & Tensilica Xtensa LX6.

Development Boards(Hands-on): Nrf52DK Boards, AVR Development Board, Arduino Uno Board,ESP32 & 8266 Board.STM32F407VG.

Protocol (Hands-on): UART, I2C, SPI, PWM, RS232, RS485, Modbus, Manchester Code, Bluetooth_Mesh.

Tools (Firmware Development): Arduino, Segger-IDE,ESP-IDF, Keil, Flash Magic, Proteus, STM32CubeIDE.

Sensors (Interfacing): Indoor Air Quality, Temperature and Relative Humidity, IR, LCD, OLED-Display, PIR, ADPD188BI, BME680.

Operating Systems: Windows, Linux (Centos, Ubuntu).

Education

BE	Shivaji University Kolhapur , Electronics & Telecommunication	2019
DIPLOMA	MSBTE Mumbai , Industrial Electronics	2012
HSC	State Board MCVC	2009

Experience

Aaroh Labs Pvt. Ltd Pune , Software Engineer	Pune FEB 2019 to Present 5 years 2 months
• Write Firmware for microcontroller based • Integrate sensors to the MCU with firmware • Debugging the software for troubleshooting • Evaluate and test the Microcontroller based product • Test and Up hardware PCB with all specification • Design Hardware for Microcontroller-specific circuits	
Aaroh Labs Pvt. Ltd Pune — Internship as a Software Engineer	FEB 2019 – July 2020
• Acquired proficiency in C programming through practical application and project work. • understanding of device driver architecture and development • Gained a deep understanding of various microcontroller and microprocessor architectures. • Demonstrated knowledge in fundamental embedded systems concepts, including real-time operating systems,memory management, and peripheral interfacing	

- Developed a strong foundation in electronics and hardware components relevant to embedded systems.
- Proficient in utilizing low-level hardware interfaces such as GPIO, SPI, I2C, and UART.
- Practical experience working with communication protocols like UART, SPI, I2C, and CAN.
- Utilized version control systems, particularly Git, for collaborative software development.

Projects:

Automated hand sanitizer dispenser

- Project 1 :- Automated hand sanitizer dispenser
- Platform:- 8051 (Microcontroller)
- Team Size: - 2 Members
- Tools :- Keil, uFlash
- **Responsibilities**
- Led the development of an automated hand sanitizer dispenser in response to limited resources during the pandemic.
- Manage included hardware design, firmware development for the microcontroller, PCB Design (third-party support)
- Ensured seamless integration of components, conducted thorough testing and provided comprehensive documentation.
- Implemented improvements based on testing feedback to enhance overall performance

Smart Smoke Detector(wireless)

- Project 2 :- Smart Smoke Detector(wireless)
- Platform :- Arm M4F,Tensilica Xtensa LX6
- Team Size: - 3 Members
- Tools:- Segger Embedded Studio, Eclipse, ESP-IDF, JLink
- **Responsibilities**
- Initiated a ground-up project involving space constraints and adherence to industry standards.
- Wrote a microcontroller-based device driver for I2C interfacing.
- - Register Configuration: Defined and implemented register settings for optimal sensor performance.
 - Startup Sequence: Created an efficient startup sequence to ensure reliable sensor initialization.
 - Interrupt Handling: Designed and implemented interrupt service routines to enhance performance and responsiveness.
- Designed and integrated various peripherals like PWM,ADC for enhanced functionality.
- Implemented power optimization strategies for battery-operated devices. comprehensive documentation.
- Spearheaded the incorporation of a Bluetooth mesh stack for seamless connectivity.
- Led debugging and testing efforts to ensure robust performance.
- Bluetooth mesh Provisioning process for Forming mesh network

Smoke Detector(Wired)

- Project 3 :- Smoke Detector(Wired)
- Platform :- Cortex-M0,Cortex-M4F
- Team Size: - 4 Members
- Tools:- CCS, JLink
- **Responsibilities**
- Created a UART driver for debugging logs on the console
- Implemented Manchester coding technique for reliable data transmission and reception in various projects.
- Developed and maintained interfaces for multiple sensors using I2C and SPI protocols, ensuring efficient communication and data acquisition.
- Contributed to designing and reviewing circuit schematics and PCB layouts, ensuring optimal component selection and functionality standards.
- Interfaced the smoke sensor and other essential components, ensuring seamless operation within the system.
- Conducted extensive testing and validation to ensure the reliability and accuracy of the smoke detector. comprehensive documentation.

Consultation Projects: _____

Home Automation

- Project 1 :- Home Automation
- Platform:- ESP32
- Team Size: - 1 Member
- Tools :- ESP-IDF,ESP-Rainmaker

Automatic wireless charging station for vehicles

- Project 2 :- Home Automation
- Platform:- Arduino
- Team Size: - 1 Member
- Tools :- Arduino IDE

PERSONAL DETAILS: _____

DOB: 10 April 1991

Nationality: INDIAN

Gender: Male

Permanent Add: Sunshine Terrace B Building Flat No-14 Sangam Nager Old Sangavi pune Pin - 411027

Languages: Marathi,Hindi,English

DECLARATION: _____

I Praful Jayant Patil hereby declare that the above information supplied is correct to the best of my knowledge.

Date:

Place: Pune

Praful Jayant Patil